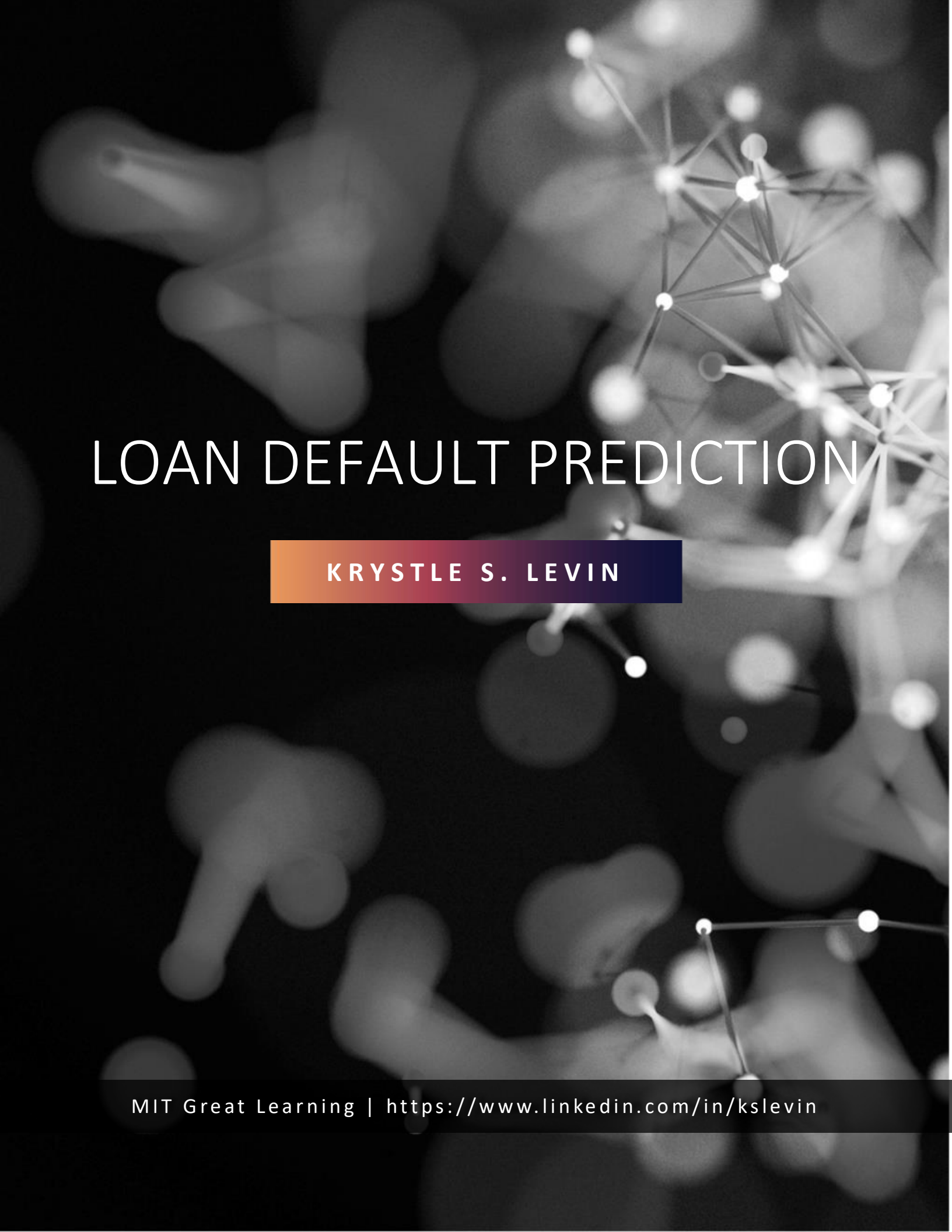




LOAN DEFAULT PREDICTION

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SUMMARY

This project proposes the Decision Tree classification model for predicting the outcome of home equity loans. It was suggested that bias existed within the dataset and that the machine should be trained to ignore them. The built model has sufficient accuracy with attention to reduction of false negatives, lowering the number of approved loans that result in default. Despite this, the model was still subject to limitations mainly stemming from the quantity of missing values. It is recommended that lenders consider requiring inclusion of all applicant information as a requirement prior to sending the application for review and that all incomplete applications should be denied.

INTRODUCTION

PROBLEM STATEMENT

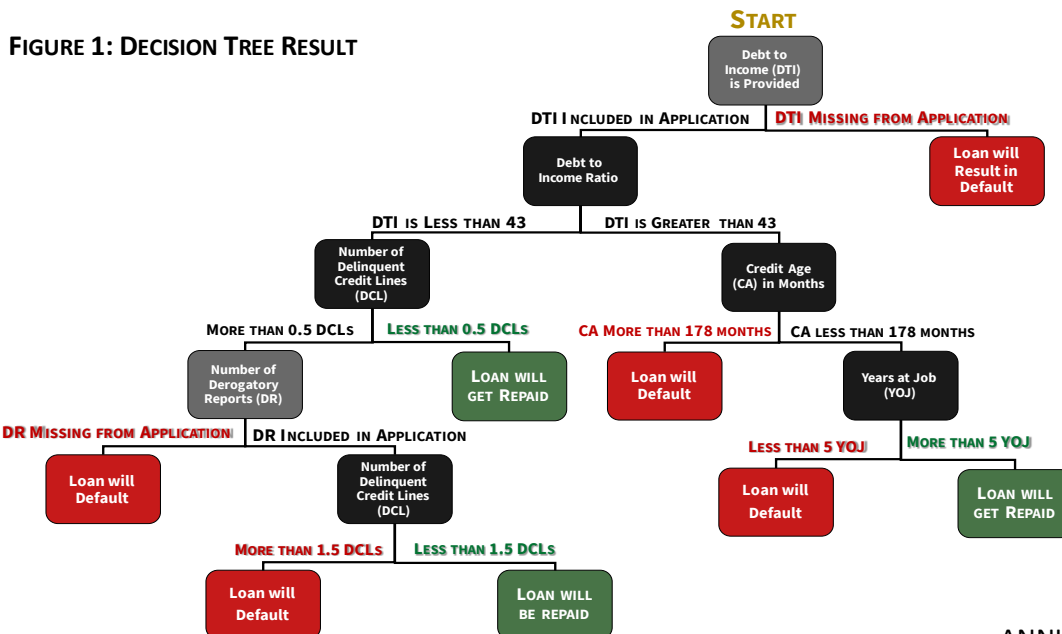
The main proponent for retail bank profits come from interest in the form of home loans. Loans of these types ending in default can present a significant loss to a lender. With defaulters making up 20% the risk of loss for home loans has potential to outweigh the potential profit received in the form of interest on these loans.

Historically, the process for reviewing loan applications and credit worthiness of an applicant has been dependent on manual and labor-intensive study of the information included in the application. Not only does this process require more staff and resources to accomplish, but it also opens the approval process up to human error and bias.

As a result, banks require a machine learning model that will simplify the process, more accurately determine credit worthiness, and be free of bias in such a way that complies with the Equal Credit Opportunity Act's guidelines and Lender decisions can be Justified. To find a solution for this, different classification models were developed with a focus on reducing the number approved loans that would ultimately end in default. The final model will serve as a basis for loan approvals and business recommendations to further reduce these false negatives and better reduce the risk they have to lenders.

SOLUTION DESIGN

In determining the best model solution for the problem, three classification models were developed, tuned, and compared. Among Logistic Regression, Decision Tree, and Random Forrest, a tuned Decision Tree was found to best address the needs of the lenders. This model, once tuned, gave sufficient accuracy and produced a white box model that is easily interpretable, as shown in **Figure 1**. While more automated processes are being built out for the loan decision process, this model can easily be used by loan reviewers to make a simplified decision on an application. Moreover, the significance of these can easily be tested using traditional statistical means in cases where decision justification is required.





ANALYSIS AND INSIGHTS

During the development of the models several discrepancies and influencing features of a loan applicant were identified as having a significant impact on the loan outcome.

MISSING DATA VALUES

Missing values played a significant role in the prediction of a loan outcome. Debt-to-Income was missing from 21% from the data points and was found that when missing, the applicant had only less than 40% chance of repaying their loan. Of the 20% of loans that defaulted, over half of them could have been avoided had applicants without this information been denied. Additionally, the lack of disclosure for the number of derogatory reports, which indicate serious delinquency or late payments, was also found to be a moderate factor in predicting loan outcomes. Lenders should evaluate if the omittance of these fields was intentional to get through the approval process and implement policies or thresholds accordingly.

SIGNIFICANT FEATURES

When Debt-to-Income ratios were included in applications, it was found that the ratio of an applicant's gross monthly income to their monthly debt payments, when higher than 43, were significantly more at risk for defaulting, unless the age of their oldest credit line was less than 178 months and they had more than 5 years at their current job. **Figure 2** shows the outcome of borrower loans based on their debt-to-income ratio, where blue (1) indicates a defaulted loan. In borrowers that had a Debt-to-Income lower than 43, those with less than 0.5 delinquent credit lines had a high chance of repaying their loan. Of those that didn't, if the number of derogatory reports was included, the threshold of tolerance on the number of derogatory reports for an applicant was raised to 1.5. If the number of derogatory reports was omitted or they had more than 1.5 delinquent credit lines, the model predicts that borrower to end in default. The correlation of the amount of derogatory reports to loan outcomes can be clearly seen in **Figure 3**, where there is a significantly less blue (1) loans that ended in default for those within the aforementioned ranges, as compared to those above the identified threshold.

FIGURE 2: LOAN OUTCOME BY DEBT-TO-INCOME

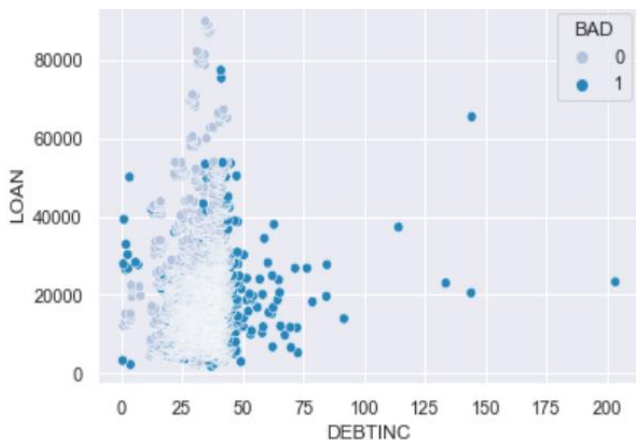
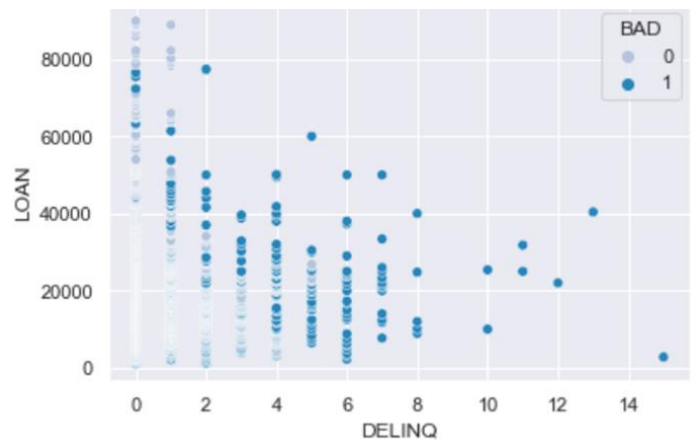


FIGURE 3: LOAN OUTCOME BY COUNT OF DEROGATORY REPORTS





BUSINESS PRIORITIES

- Reduce the Number of Approved Loans that End in Default
- Simplified Decision Process
- Bias-Free Approvals and Rejections
- Interpretable Model that can easily be leveraged to justify final decisions



Key Action Items

- Application Completion Requirements
- Defined Thresholds Requirements for Debt-to-Income Ratios and Acceptable Quantities of Derogatory Reports
- Additional Income and Debt Granularity

BUISSINESS RECCOMENDATIONS

IMMEDIATE ACTIONS

To start, lenders should make completion of the application a requirement before being sent for review. The significance of the missing values indicates the lack of policy or enforcement around this, and it has proven to have a significant impact on the prediction of loan repayment or default. Secondly, lenders need to set threshold limits for debt-to-income ratios, and then dependent thresholds for acceptable number of derogatory reports, a minimum credit age, and a minimum number of years at an applicant's current job, based on subsequent information.

LONG TERM SOLUTIONS

To fully comply with the Equal Credit Opportunity Act's guidelines for an empirically derived and statistically sound model for credit scoring, retail bank should invest in automation tools that automate the decision process based on the applicant's details further eliminate the bias and human error caused by human approvals. Additionally, investments in support for borrowers, such as financial advisors, financial planning courses, and payment deferment programs can help reduce the likelihood of at risk approved borrowers from having their loan end in default.

MODEL BENEFITS AND RISKS

While the model did well in reducing the number of approved borrowers that ended in default, it was not perfect in its rejection and rejected applications that would have resulted in repayment. The model did significantly reduce the number of approved loans that ended in default by about 75%. However, of those that would have been repaid about 34% of them were denied by the model. That being said, the profit on the interest from the 10% is not likely to outweigh the 75% reduction in loss from approving loans that end in default.

CHALLENGES

The missing values, the lack of variation in the data set, and human error will continue to be significant contributors in the performance of the model if not addressed. Capturing additional data points, such as an applicant's income and overall debt separately, can not only help in increasing the granularity of the analysis, but also help providing validation for other fields, such as debt to income, to cross check against human error and bias.

APPENDIX:

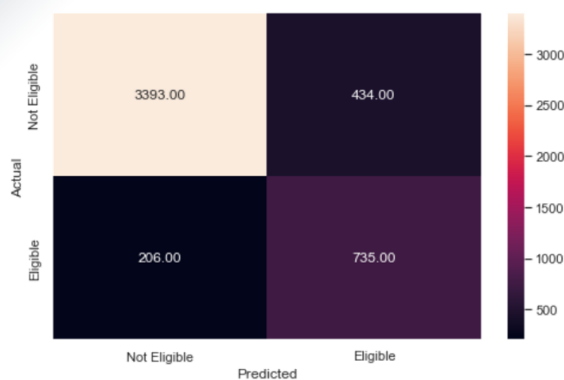
Table 1: Model Performance Comparison

		Logistic Regression	Decision Tree	Tuned Decision Tree	Random Forest	Weighted Random Forest	Tuned Random Forrest
Train	Accuracy	0.80	1.0	0.86	1.0	1.0	0.89
	Recall	0.34	1.0	0.78	1.0	1.0	0.81
	Precision	0.71	1.0	0.63	1.0	1.0	0.69
Test	Accuracy	0.79	0.87	0.86	0.91	0.91	0.87
	Recall	0.01	0.63	0.73	0.69	0.67	0.73
	Precision	0.50	0.71	0.65	0.87	0.87	0.69

Figure 4: Tuned Decision Tree Performance

Training Set

	precision	recall	f1-score	support
0	0.94	0.89	0.91	3827
1	0.63	0.78	0.70	941
accuracy			0.87	4768
macro avg	0.79	0.83	0.81	4768
weighted avg	0.88	0.87	0.87	4768



Testing Set

	precision	recall	f1-score	support
0	0.93	0.90	0.91	944
1	0.65	0.73	0.69	248
accuracy			0.86	1192
macro avg	0.79	0.81	0.80	1192
weighted avg	0.87	0.86	0.87	1192



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Figure 5: Feature Importance of Decision Tree

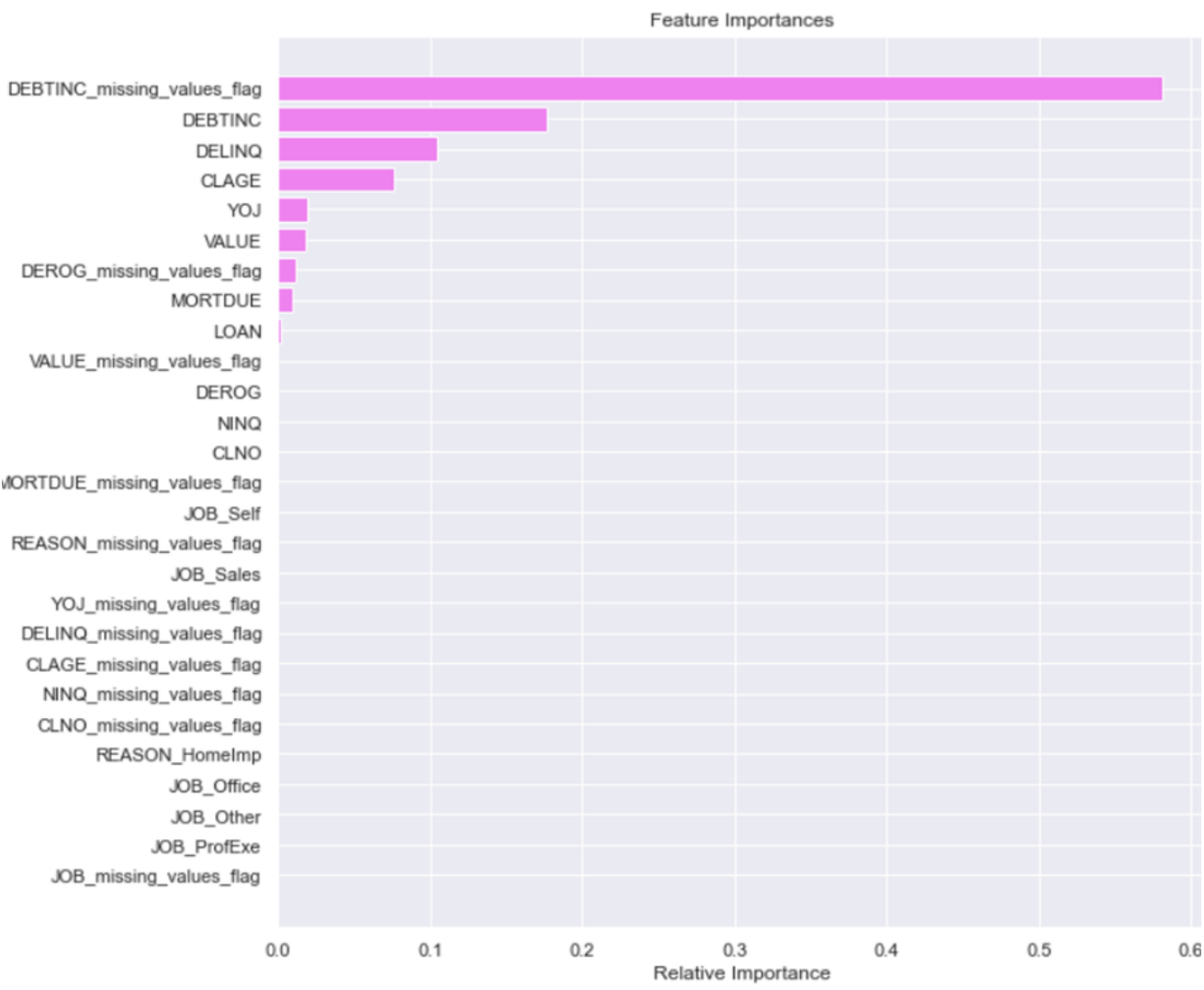


Table 2: Probabilities of 4 Most Important Features

Feature Metric	# of Occurrences	P(Default, Given the Feature has Occurred)
DEBTINC is missing	1267	60%
DEBTINC is greater than 44	103	85%
CLAGE is less than 288 Months	5037	21%
DELINQ is greater than 1.5	547	57%
Two Feature Metrics	# of Occurrences	P(Default, Given Both Features Occurred)
DEBTINC > 44 and CLAGE < 288	96	89%
DEBTINC > 44 and CLAGE < 288	19	100%

Figure 7: Decision Tree Model Output

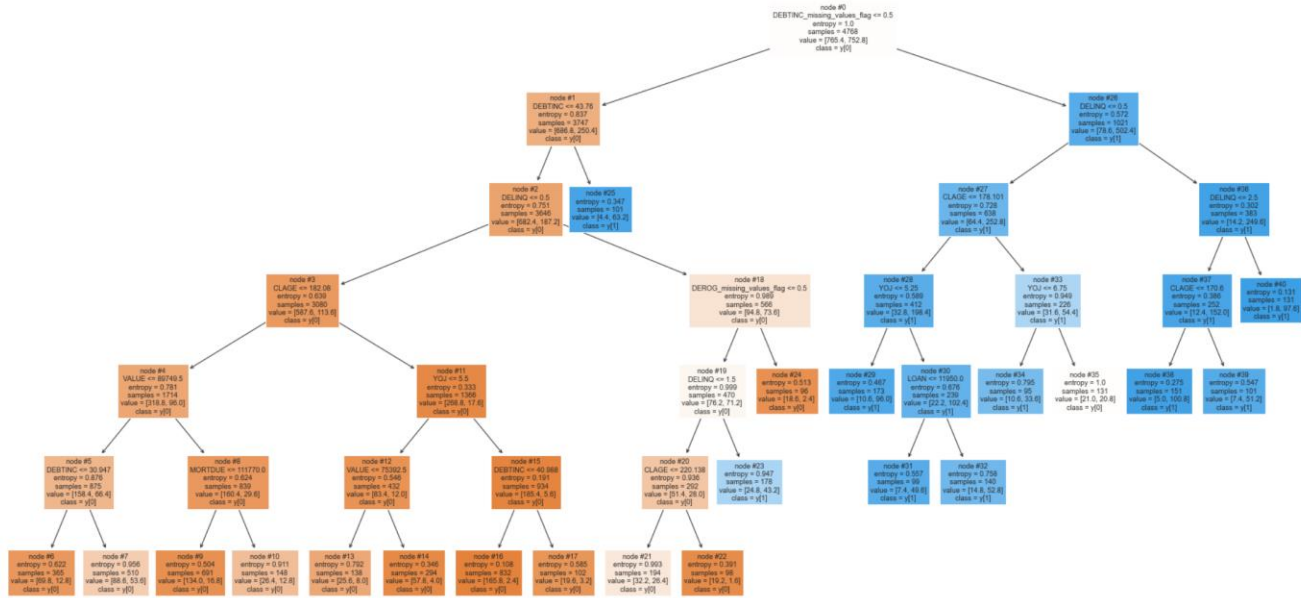


Figure 8: Decision Tree Code Highlights

```
In [184]: 1 # Choose the type of classifier.
2
3 d_tree_tuned = DecisionTreeClassifier(random_state = 5, class_weight = {0: 0.2, 1: 0.8})
4
5
6 # Grid of parameters to choose from
7
8 parameters = {'splitter': ['best', 'random'],
9               'max_depth': np.arange(3,8),
10              'criterion': ['gini', 'entropy'],
11              'min_samples_leaf': np.arange(0,100,5),
12              'min_impurity_decrease' : [0,.1,.2,.5]
13             }
14
15 # Type of scoring used to compare parameter combinations
16
17 scorer = metrics.make_scorer(recall_score, pos_label = 1)
18
19
20 # Run the grid search
21 grid_obj = GridSearchCV(d_tree_tuned, parameters, scoring = scorer, cv = 5)
22
23
24 # Fit the GridSearch on train dataset
25 grid_obj = grid_obj.fit(X_train, y_train)
26
27 # Set the clf to the best combination of parameters
28 d_tree_tuned = grid_obj.best_estimator_
29
30
31 # Fit the best algorithm to the data.
32 d_tree_tuned.fit(X_train, y_train)
```

```
Out[184]: DecisionTreeClassifier(class_weight={0: 0.2, 1: 0.8}, criterion='entropy',
                                max_depth=6, min_impurity_decrease=0,
                                min_samples_leaf=95, random_state=5)
```


Figure 9: NULL Value Distribution Analysis

